# SF2943 Part A — Full Pipeline

## Setup

Observations $X_1, \ldots, X_n$ with $X_t > 0$ the daily mean electricity load in MW at day index $t$, $n = 2100$. Goal: forecasts $\hat{X}_{n+1}, \ldots, \hat{X}_{n+H}$ with approximate 95% prediction intervals, $H = 30$.

The series is not weakly stationary: $\mathbb{E}[X_t]$ depends on $t$ (long-run trend, weekly cycle, yearly cycle) and $\mathrm{Var}(X_t)$ grows with $\mathbb{E}[X_t]$.

**Weak stationarity (target property).** A process $\{Y_t\}_{t\in\mathbb{Z}}$ is weakly stationary if

$$\mathbb{E}[Y_t] = \mu \quad \text{(constant in } t\text{)}, \qquad \mathrm{Cov}(Y_t, Y_{t+h}) = \gamma(h) \quad \text{(depends only on lag } h\text{)}.$$

Here $\mu$ is a single number, $\gamma : \mathbb{Z} \to \mathbb{R}$ is the *autocovariance function*, $\gamma(0) = \mathrm{Var}(Y_t)$, and the *autocorrelation function* is $\rho(h) = \gamma(h)/\gamma(0)$.

**Pipeline.**

**Step 1.** Clean: transform $X_t$ into a residual $\hat{Y}_t$ that is approximately weakly stationary.

**Step 2.** Fit: pick a parametric ARMA$(p, q)$ model and estimate its parameters from $\hat{Y}_t$.

**Step 3.** Forecast: use the fitted model to predict $Y_{n+h}$ for $h = 1, \ldots, H$, then invert the cleaning to return to MW.

## 1 Cleaning

### 1.1 Decomposition model

Posit

$$\log X_t \;=\; m_t \;+\; s_t^{(\mathrm{year})} \;+\; s_t^{(\mathrm{week})} \;+\; Y_t.$$

Components:

- $m_t$: deterministic trend (slowly varying function of $t$).
- $s_t^{(\mathrm{year})}$: deterministic seasonal of period $d_y = 365$, satisfying $s_{t-d_y}^{(\mathrm{year})} = s_t^{(\mathrm{year})}$ and $\sum_{t=1}^{d_y} s_t^{(\mathrm{year})} = 0$.
- $s_t^{(\mathrm{week})}$: deterministic seasonal of period $d_w = 7$, analogous.
- $Y_t$: residual. Target property: weakly stationary.

The job of Step 1 is to estimate $\hat{m}_t, \hat{s}_t^{(\mathrm{year})}, \hat{s}_t^{(\mathrm{week})}$ and form $\hat{Y}_t = \log X_t - \hat{m}_t - \hat{s}_t^{(\mathrm{year})} - \hat{s}_t^{(\mathrm{week})}$.

### 1.2 Why log first

Suppose

$$\mathrm{Var}(X_t) \;\approx\; c\,\mu_X(t)^2, \qquad \mu_X(t) := \mathbb{E}[X_t],$$

for some constant $c > 0$. We seek a smooth $g : (0, \infty) \to \mathbb{R}$ such that $\mathrm{Var}(g(X_t))$ does not depend on $t$.

Linearize $g$ around $\mu := \mu_X(t)$ (delta method):

$$g(X_t) \;\approx\; g(\mu) + g'(\mu)\,(X_t - \mu).$$

The first term is deterministic at fixed $t$, contributes nothing to variance. Taking variance:

$$\mathrm{Var}(g(X_t)) \;\approx\; [g'(\mu)]^2 \cdot \mathrm{Var}(X_t) \;\approx\; [g'(\mu)]^2 \cdot c\,\mu^2.$$

For this to be constant in $t$ (independent of $\mu$), require

$$[g'(\mu)]^2 \cdot \mu^2 = \text{const} \iff g'(\mu) = \frac{k}{\mu} \iff g(\mu) = k \log \mu + \text{const}.$$

Drop constants: $g = \log$. This is the Box–Cox transform with $\lambda = 0$.

**Bonus.** If the structure is $X_t = L_t \cdot S_t \cdot N_t$ (level $\times$ seasonal $\times$ noise, multiplicative), then $\log X_t = \log L_t + \log S_t + \log N_t$ (additive), matching the decomposition model.

### 1.3 Yearly seasonality: harmonic regression

Parameterize $s_t^{(\text{year})}$ as a truncated Fourier series with $k_y = 2$ harmonics:

$$s_t^{(\text{year})} \;=\; \sum_{j=1}^{k_y} \Big[ a_j^{(y)} \cos\big(\lambda_j^{(y)} t\big) + b_j^{(y)} \sin\big(\lambda_j^{(y)} t\big) \Big], \qquad \lambda_j^{(y)} = \frac{2\pi j}{d_y} = \frac{2\pi j}{365}.$$

Variables: $j$ is the harmonic index, $\lambda_j^{(y)}$ is the angular frequency in radians per day, and $a_j^{(y)}, b_j^{(y)}$ are unknown coefficients (4 in total for $k_y = 2$).

**Why this is automatically periodic with period $d_y$.** For any integer $j$,

$$\cos\big(\lambda_j^{(y)}(t + d_y)\big) = \cos\big(\tfrac{2\pi j t}{d_y} + 2\pi j\big) = \cos\big(\lambda_j^{(y)} t\big),$$

since $2\pi j$ is a full integer number of rotations. Same for sine.

**Why $k_y$ harmonics give $2k_y$ free parameters.** Each harmonic contributes one cosine and one sine. The pair $(a_j, b_j)$ is equivalent to amplitude–phase $(R_j, \varphi_j)$ via $a_j \cos(\lambda t) + b_j \sin(\lambda t) = R_j \cos(\lambda t - \varphi_j)$ with $R_j = \sqrt{a_j^2 + b_j^2}$ and $\varphi_j = \arctan(b_j/a_j)$, but the $(a_j, b_j)$ form is linear in the parameters and the $(R_j, \varphi_j)$ form is not. We fit in $(a_j, b_j)$ and convert afterwards if needed.

**Estimation by ordinary least squares.** Define

$$\mathbf{x} := (\log X_1, \log X_2, \ldots, \log X_n)^\top \in \mathbb{R}^n, \qquad \boldsymbol{\beta}_y := (a_1^{(y)}, b_1^{(y)}, a_2^{(y)}, b_2^{(y)})^\top \in \mathbb{R}^{2k_y}.$$

Build the design matrix

$$A_y \;=\; \begin{pmatrix} \cos(\lambda_1^{(y)} \cdot 1) & \sin(\lambda_1^{(y)} \cdot 1) & \cos(\lambda_2^{(y)} \cdot 1) & \sin(\lambda_2^{(y)} \cdot 1) \\ \cos(\lambda_1^{(y)} \cdot 2) & \sin(\lambda_1^{(y)} \cdot 2) & \cos(\lambda_2^{(y)} \cdot 2) & \sin(\lambda_2^{(y)} \cdot 2) \\ \vdots & \vdots & \vdots & \vdots \\ \cos(\lambda_1^{(y)} \cdot n) & \sin(\lambda_1^{(y)} \cdot n) & \cos(\lambda_2^{(y)} \cdot n) & \sin(\lambda_2^{(y)} \cdot n) \end{pmatrix} \in \mathbb{R}^{n \times 2k_y}.$$

The model is $\mathbf{x} \approx A_y \boldsymbol{\beta}_y$. Minimize the squared residual:

$$S(\boldsymbol{\beta}_y) \;=\; \|\mathbf{x} - A_y \boldsymbol{\beta}_y\|^2 \;=\; \sum_{t=1}^{n} \left( \log X_t - \sum_{\ell=1}^{2k_y} (A_y)_{t\ell}\, (\boldsymbol{\beta}_y)_\ell \right)^2.$$

Set $\nabla_{\boldsymbol{\beta}_y} S = 0$:

$$-2A_y^\top(\mathbf{x} - A_y\boldsymbol{\beta}_y) = 0 \implies A_y^\top A_y\,\boldsymbol{\beta}_y = A_y^\top \mathbf{x} \implies \boxed{\hat{\boldsymbol{\beta}}_y = (A_y^\top A_y)^{-1} A_y^\top \mathbf{x}.}$$

Form $\hat{\mathbf{s}}^{(\text{year})} = A_y\hat{\boldsymbol{\beta}}_y \in \mathbb{R}^n$ and update:

$$\mathbf{x}^{(1)} := \mathbf{x} - \hat{\mathbf{s}}^{(\text{year})}.$$

### 1.4 Weekly seasonality

Same procedure on $\mathbf{x}^{(1)}$ with $d_w = 7$, $k_w = 3$:

$$s_t^{(\text{week})} \;=\; \sum_{j=1}^{k_w}\Big[a_j^{(w)}\cos\big(\lambda_j^{(w)}t\big) + b_j^{(w)}\sin\big(\lambda_j^{(w)}t\big)\Big], \qquad \lambda_j^{(w)} = \frac{2\pi j}{7}.$$

Build $A_w \in \mathbb{R}^{n\times 2k_w} = \mathbb{R}^{n\times 6}$ analogously, solve

$$\hat{\boldsymbol{\beta}}_w = (A_w^\top A_w)^{-1} A_w^\top \mathbf{x}^{(1)},$$

form $\hat{\mathbf{s}}^{(\text{week})} = A_w\hat{\boldsymbol{\beta}}_w$, update:

$$\mathbf{x}^{(2)} := \mathbf{x}^{(1)} - \hat{\mathbf{s}}^{(\text{week})}.$$

### 1.5 Polynomial trend

Try linear first. Define $\mathbf{t} = (1, 2, \ldots, n)^\top$, build

$$A_m^{(1)} = [\,\mathbf{1}\;\;\mathbf{t}\,] \in \mathbb{R}^{n\times 2}, \qquad \boldsymbol{\beta}_m^{(1)} = (\beta_0, \beta_1)^\top.$$

Solve $\hat{\boldsymbol{\beta}}_m^{(1)} = (A_m^{(1)\top} A_m^{(1)})^{-1} A_m^{(1)\top}\mathbf{x}^{(2)}$, form $\hat{\mathbf{m}}^{(1)} = A_m^{(1)}\hat{\boldsymbol{\beta}}_m^{(1)}$, residuals $\mathbf{r}^{(1)} = \mathbf{x}^{(2)} - \hat{\mathbf{m}}^{(1)}$.

**Curvature check.** Split $\mathbf{r}^{(1)}$ into thirds $\mathbf{r}_{[1]}^{(1)}, \mathbf{r}_{[2]}^{(1)}, \mathbf{r}_{[3]}^{(1)}$ of $n/3$ entries each, compute group means $\bar{r}_i$ and overall std $\hat{\sigma}_r$. If

$$\max_{i\in\{1,2,3\}} |\bar{r}_i| \;>\; 0.10\,\hat{\sigma}_r$$

then the linear trend left systematic curvature; refit with quadratic basis $A_m^{(2)} = [\,\mathbf{1}\;\;\mathbf{t}\;\;\mathbf{t}^2\,] \in \mathbb{R}^{n\times 3}$ to obtain $\hat{\mathbf{m}}^{(2)}$. The final $\hat{\mathbf{m}}$ is whichever was chosen.

### 1.6 Residual series

$$\boxed{\hat{Y}_t \;:=\; \log X_t - \hat{m}_t - \hat{s}_t^{(\text{year})} - \hat{s}_t^{(\text{week})}, \qquad t = 1, \ldots, n.}$$

### 1.7 Diagnostics

**Sample autocovariance and autocorrelation.**

$$\hat{\gamma}(h) \;=\; \frac{1}{n}\sum_{t=1}^{n-|h|}\big(\hat{Y}_{t+|h|} - \bar{\hat{Y}}\big)\big(\hat{Y}_t - \bar{\hat{Y}}\big), \qquad \hat{\rho}(h) \;=\; \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)},$$

with $\bar{\hat{Y}} = \frac{1}{n}\sum_{t=1}^n \hat{Y}_t$. Divisor $n$ (rather than $n - |h|$) ensures the sample covariance matrix $\hat{\Gamma}_n = [\hat{\gamma}(i-j)]_{i,j=1}^n$ is nonnegative definite.

**Visual check.** For an iid sequence with finite variance,

$$\hat{\rho}(h) \;\overset{\text{approx}}{\sim}\; \mathcal{N}\big(0, \tfrac{1}{n}\big) \quad \text{for } h \neq 0,$$

so under iid approximately 95% of the $\hat{\rho}(h)$ lie in $\left[-1.96/\sqrt{n},\, +1.96/\sqrt{n}\right]$. Plot $\hat{\rho}(h)$ with these bounds. Cleaning succeeded if there is no slow geometric decay (would indicate residual trend) and no spikes at $h = 7$ or $h = 365$ (would indicate residual seasonality).

**Ljung–Box portmanteau test.**

$$Q_{\mathrm{LB}} \;=\; n(n+2)\sum_{j=1}^{h} \frac{\hat{\rho}^2(j)}{n-j}.$$

Under the null $H_0 : \{\hat{Y}_t\}$ iid, $Q_{\mathrm{LB}} \overset{\text{approx}}{\sim} \chi^2(h)$. Reject $H_0$ if $Q_{\mathrm{LB}} > \chi^2_{0.95}(h)$. At this stage we *want* to reject: rejection means there is correlation structure left for ARMA to model.

# 2 ARMA fitting

## 2.1 Model class

A causal ARMA$(p, q)$ process satisfies

$$Y_t \;=\; \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} \;+\; Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \cdots + \theta_q Z_{t-q},$$

$$\{Z_t\} \sim \mathrm{WN}(0, \sigma^2),$$

where $\{Z_t\}$ are uncorrelated zero-mean random variables with constant variance $\sigma^2$ (the *innovations* or *shocks*). Define the backshift operator $B$ by $BY_t = Y_{t-1}$ and the polynomials

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p, \qquad \theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q.$$

Compact form:

$$\phi(B)Y_t \;=\; \theta(B)Z_t.$$

Parameter vector:

$$\boldsymbol{\beta} \;=\; (\phi_1, \ldots, \phi_p,\; \theta_1, \ldots, \theta_q,\; \sigma^2) \;\in\; \mathbb{R}^{p+q+1}.$$

Total parameter count $k = p + q + 1$.

**Causality.** The ARMA process is *causal* if all roots $z_i$ of $\phi(z) = 0$ satisfy $|z_i| > 1$. Equivalently, $Y_t$ admits an absolutely convergent representation as a one-sided MA in present and past shocks:

$$Y_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \qquad \sum_{j=0}^{\infty} |\psi_j| < \infty.$$

**Invertibility.** The process is *invertible* if all roots $z_i$ of $\theta(z) = 0$ satisfy $|z_i| > 1$. Equivalently, $Z_t$ admits an absolutely convergent representation as a one-sided AR in present and past observations:

$$Z_t = \sum_{j=0}^{\infty} \pi_j Y_{t-j}, \qquad \sum_{j=0}^{\infty} |\pi_j| < \infty.$$

Both conditions are required for estimation and prediction to be well-defined.

## 2.2 Identification: sample PACF

The sample partial autocorrelation function $\hat{\alpha}(\cdot)$ is defined recursively as the deepest coefficient in fitted AR models of increasing order. At lag $k$, fit

$$\hat{Y}_t \;=\; \phi_{k1}\hat{Y}_{t-1} + \phi_{k2}\hat{Y}_{t-2} + \cdots + \phi_{kk}\hat{Y}_{t-k} + \text{error}$$

by ordinary least squares (equivalently sample Yule–Walker, see §2.3 below) and set

$$\hat{\alpha}(k) \;:=\; \hat{\phi}_{kk}.$$

In matrix form: with $\hat{\Gamma}_k = [\hat{\gamma}(i-j)]_{i,j=1}^{k}$ and $\hat{\boldsymbol{\gamma}}_k = (\hat{\gamma}(1), \ldots, \hat{\gamma}(k))^\top$,

$$\hat{\boldsymbol{\phi}}^{(k)} \;=\; \hat{\Gamma}_k^{-1}\,\hat{\boldsymbol{\gamma}}_k, \qquad \hat{\alpha}(k) = \big(\hat{\boldsymbol{\phi}}^{(k)}\big)_k.$$

**Diagnostic table.**

| Process | sample ACF $\hat{\rho}(h)$ | sample PACF $\hat{\alpha}(k)$ |
|---|---|---|
| AR($p$) | tails off | cuts off after lag $p$ |
| MA($q$) | cuts off after lag $q$ | tails off |
| ARMA($p,q$) | tails off | tails off |

"Cuts off" means: drops inside $\pm 1.96/\sqrt{n}$ and stays. Read both shapes from data and list 3–5 candidate orders $(p_i, q_i)$.

## 2.3 Yule–Walker estimation (pure AR, used for cross-check)

Assume $q = 0$, so $Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t$. Multiply both sides by $Y_{t-k}$ for $k = 1, \ldots, p$ and take expectations:

$$\mathbb{E}[Y_t Y_{t-k}] \;=\; \sum_{i=1}^{p} \phi_i\, \mathbb{E}[Y_{t-i} Y_{t-k}] + \mathbb{E}[Z_t Y_{t-k}].$$

By stationarity $\mathbb{E}[Y_t Y_{t-k}] = \gamma(k)$. Since $Z_t$ is uncorrelated with $Y_{t-k}$ for $k \geq 1$, the last term vanishes:

$$\gamma(k) \;=\; \sum_{i=1}^{p} \phi_i\, \gamma(k-i), \qquad k = 1, \ldots, p.$$

Stack into matrix form with

$$\Gamma_p = \begin{pmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(p-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(p-2) \\ \vdots & & \ddots & \vdots \\ \gamma(p-1) & \gamma(p-2) & \cdots & \gamma(0) \end{pmatrix}, \quad \boldsymbol{\gamma}_p = \begin{pmatrix} \gamma(1) \\ \gamma(2) \\ \vdots \\ \gamma(p) \end{pmatrix}, \quad \boldsymbol{\phi} = \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix}.$$

$$\boxed{\Gamma_p\,\boldsymbol{\phi} \;=\; \boldsymbol{\gamma}_p, \qquad \sigma^2 \;=\; \gamma(0) - \boldsymbol{\phi}^\top \boldsymbol{\gamma}_p.}$$

The sample version replaces $\gamma$ with $\hat{\gamma}$:

$$\hat{\boldsymbol{\phi}} \;=\; \hat{\Gamma}_p^{-1}\,\hat{\boldsymbol{\gamma}}_p, \qquad \hat{\sigma}^2 \;=\; \hat{\gamma}(0) - \hat{\boldsymbol{\phi}}^\top \hat{\boldsymbol{\gamma}}_p.$$

Closed form, no iteration. Limitation: only handles $q = 0$. Used in this project for the PACF computation (via the Durbin–Levinson recursion at orders $1, 2, \ldots$) and as a cross-check on the best AR candidate against the MLE estimate.

## 2.4 Gaussian maximum likelihood estimation

Assume $Z_t \sim \mathcal{N}(0, \sigma^2)$ iid. Then $\mathbf{Y} = (Y_1, \ldots, Y_n)^\top$ is jointly Gaussian with mean $\mathbf{0}$ and covariance matrix

$$\Gamma_n(\boldsymbol{\beta}) = [\gamma(i-j;\, \boldsymbol{\beta})]_{i,j=1}^{n},$$

where $\gamma(\,\cdot\,;\,\boldsymbol{\beta})$ is the autocovariance function determined by the ARMA parameters $\boldsymbol{\beta}$. Joint density:

$$f(\mathbf{y};\boldsymbol{\beta}) \;=\; (2\pi)^{-n/2}\,\big|\Gamma_n(\boldsymbol{\beta})\big|^{-1/2}\exp\Big(-\tfrac{1}{2}\,\mathbf{y}^\top\Gamma_n(\boldsymbol{\beta})^{-1}\mathbf{y}\Big)\,.$$

Direct evaluation requires inverting an $n\times n$ matrix, which is impractical for $n=2100$. Avoid by chain-rule factorization:

$$f(y_1,\ldots,y_n) \;=\; f(y_1)\prod_{j=2}^{n} f(y_j\mid y_1,\ldots,y_{j-1}).$$

Each conditional $f(y_j\mid y_1,\ldots,y_{j-1})$ is univariate Gaussian with

$$\text{mean} = \hat{Y}_j := \text{best linear predictor of } Y_j \text{ from } Y_1,\ldots,Y_{j-1},$$

$$\text{variance} = v_{j-1} := \mathbb{E}\Big[(Y_j-\hat{Y}_j)^2\Big]\,.$$

Both depend on $\boldsymbol{\beta}$ through $\gamma(\,\cdot\,;\,\boldsymbol{\beta})$. Hence

$$f(y_j\mid y_1,\ldots,y_{j-1}) \;=\; \frac{1}{\sqrt{2\pi\,v_{j-1}}}\,\exp\left(-\frac{(y_j-\hat{Y}_j)^2}{2v_{j-1}}\right).$$

Multiplying and taking logs:

$$\boxed{\ln L(\boldsymbol{\beta}) \;=\; -\frac{n}{2}\ln(2\pi) \;-\; \frac{1}{2}\sum_{j=1}^{n}\ln v_{j-1}(\boldsymbol{\beta}) \;-\; \frac{1}{2}\sum_{j=1}^{n}\frac{\big(\hat{y}_j-\hat{Y}_j(\boldsymbol{\beta})\big)^2}{v_{j-1}(\boldsymbol{\beta})}.}$$

The data $\hat{y}_j$ are fixed; $\boldsymbol{\beta}$ enters through $\hat{Y}_j(\boldsymbol{\beta})$ and $v_{j-1}(\boldsymbol{\beta})$. The MLE is

$$\hat{\boldsymbol{\beta}}_{\text{MLE}} \;=\; \arg\max_{\boldsymbol{\beta}}\,\ln L(\boldsymbol{\beta}).$$

The system $\nabla_{\boldsymbol{\beta}}\ln L = 0$ has no closed-form solution because $\hat{Y}_j$ and $v_{j-1}$ are produced by a recursion in $\boldsymbol{\beta}$, not closed-form expressions. Solve numerically.

## 2.5 Innovations algorithm

Computes $\hat{Y}_j$ and $v_{j-1}$ recursively from $\gamma(\,\cdot\,;\,\boldsymbol{\beta})$, in $O(n)$ work per likelihood evaluation for ARMA models.

**Initialization.**

$$\hat{Y}_1 \;=\; 0, \qquad v_0 \;=\; \gamma(0).$$

**Recursion.** For $n=1,2,\ldots,N-1$ and $k=0,1,\ldots,n-1$:

$$\theta_{n,\,n-k} \;=\; \frac{1}{v_k}\left(\gamma(n-k) \;-\; \sum_{j=0}^{k-1}\theta_{k,\,k-j}\,\theta_{n,\,n-j}\,v_j\right).$$

Then

$$v_n \;=\; \gamma(0) \;-\; \sum_{j=0}^{n-1}\theta_{n,\,n-j}^2\,v_j,$$

and the next predictor uses past innovations $\hat{y}_i-\hat{Y}_i$:

$$\hat{Y}_{n+1} \;=\; \sum_{j=1}^{n}\theta_{n,j}\,\big(\hat{y}_{n+1-j}-\hat{Y}_{n+1-j}\big).$$

The coefficients $\theta_{n,k}$ here are *innovations expansion coefficients*, distinct from the MA parameters $\theta_1, \ldots, \theta_q$ of the ARMA model (notational collision). The whole recursion takes $\gamma(\,\cdot\,;\,\boldsymbol{\beta})$ as input and produces $\{\hat{Y}_j(\boldsymbol{\beta}), v_{j-1}(\boldsymbol{\beta})\}_{j=1}^{N}$.

**Likelihood evaluation pipeline.**

$$\boldsymbol{\beta} \xrightarrow{\text{ARMA closed form}} \gamma(\,\cdot\,;\,\boldsymbol{\beta}) \xrightarrow{\text{innovations algorithm}} \{\hat{Y}_j, v_{j-1}\}_{j=1}^{n} \xrightarrow{\text{plug in}} \ln L(\boldsymbol{\beta}).$$

## 2.6 Optimization

Apply BFGS quasi-Newton starting from a Yule–Walker preliminary estimate on the AR part (zeros for the MA part). At iteration $k$ with current point $\boldsymbol{\beta}_k$:

1. Evaluate $\ln L(\boldsymbol{\beta}_k)$ via the innovations algorithm.
2. Estimate $\nabla \ln L(\boldsymbol{\beta}_k)$ by finite differences.
3. Form a quasi-Newton search direction using an estimated Hessian.
4. Line search along that direction to obtain $\boldsymbol{\beta}_{k+1}$.
5. Stop if $\|\nabla \ln L\|$ or the step size is below tolerance.

The search is constrained to the causal/invertible region of $\boldsymbol{\beta}$-space. Output: $\hat{\boldsymbol{\beta}}_{\text{MLE}}$.

## 2.7 Causality and invertibility verification

For each fitted candidate compute the roots of

$$\hat{\phi}(z) = 1 - \hat{\phi}_1 z - \cdots - \hat{\phi}_p z^p = 0 \quad \text{and} \quad \hat{\theta}(z) = 1 + \hat{\theta}_1 z + \cdots + \hat{\theta}_q z^q = 0$$

numerically. Verify $|z_i| > 1$ for all roots of both polynomials. Discard any candidate failing either condition.

## 2.8 Order selection: AICC

For each surviving candidate compute

$$\boxed{\text{AICC} \;=\; -2\ln L(\hat{\boldsymbol{\beta}}_{\text{MLE}}) \;+\; \frac{2k\,n}{n-k-1}, \qquad k = p+q+1.}$$

Equivalently $\text{AICC} = \text{AIC} + \dfrac{2k(k+1)}{n-k-1}$ with $\text{AIC} = -2\ln L + 2k$. The term $2k(k+1)/(n-k-1)$ is a finite-sample bias correction that vanishes as $n \to \infty$. Pick the candidate that minimizes AICC.

## 2.9 Diagnostic on chosen ARMA

Compute residuals

$$\hat{R}_t \;=\; \hat{y}_t - \hat{Y}_t(\hat{\boldsymbol{\beta}}_{\text{MLE}}), \qquad t = 1, \ldots, n,$$

where $\hat{Y}_t$ is the one-step predictor produced by the innovations algorithm at $\hat{\boldsymbol{\beta}}_{\text{MLE}}$. Compute the sample ACF of $\hat{R}_t$ and the Ljung–Box statistic with degrees-of-freedom correction:

$$Q_{\text{LB}}(\hat{R}) \;=\; n(n+2)\sum_{j=1}^{h} \frac{\hat{\rho}_R^2(j)}{n-j} \;\overset{\text{approx}}{\sim}\; \chi^2(h-p-q) \quad \text{under } H_0 : \{\hat{R}_t\} \text{ iid.}$$

The $p+q$ degrees of freedom subtracted account for parameters estimated from the same data. Reject the white-noise null if $Q_{\text{LB}} > \chi^2_{0.95}(h-p-q)$. Failure to reject is the desired outcome.

# 3 Forecasting

## 3.1 Best linear predictor

Given a zero-mean weakly stationary process $\{Y_t\}$ with autocovariance $\gamma$ and observations $Y_1, \ldots, Y_n$, predict $Y_{n+h}$ by a linear combination

$$P_n Y_{n+h} \;=\; \sum_{i=1}^{n} a_i \, Y_{n+1-i}$$

with deterministic coefficients $\mathbf{a} = (a_1, \ldots, a_n)^\top$. Choose $\mathbf{a}$ to minimize the mean squared prediction error

$$\mathrm{MSE}(\mathbf{a}) \;=\; \mathbb{E}\left[\left(Y_{n+h} - \sum_{i=1}^{n} a_i Y_{n+1-i}\right)^2\right].$$

Set $\partial \mathrm{MSE}/\partial a_k = 0$ for $k = 1, \ldots, n$:

$$-2\,\mathbb{E}\left[\left(Y_{n+h} - \sum_{i=1}^{n} a_i Y_{n+1-i}\right) Y_{n+1-k}\right] \;=\; 0.$$

Distribute the expectation, use stationarity $\mathbb{E}[Y_s Y_t] = \gamma(s-t)$:

$$\gamma(h+k-1) \;=\; \sum_{i=1}^{n} a_i \, \gamma(k-i), \qquad k = 1, \ldots, n.$$

Stack into matrix form:

$$\boxed{\Gamma_n \, \mathbf{a} \;=\; \boldsymbol{\gamma}_n(h),}$$

where

$$\Gamma_n = [\gamma(i-j)]_{i,j=1}^{n}, \qquad \boldsymbol{\gamma}_n(h) = \big(\gamma(h),\, \gamma(h+1),\, \ldots,\, \gamma(h+n-1)\big)^\top.$$

Solve:

$$\hat{\mathbf{a}} \;=\; \Gamma_n^{-1} \, \boldsymbol{\gamma}_n(h).$$

The forecast and its mean squared prediction error are

$$P_n Y_{n+h} \;=\; \hat{\mathbf{a}}^\top \mathbf{Y}, \qquad v_n(h) \;:=\; \mathbb{E}\big[(Y_{n+h} - P_n Y_{n+h})^2\big] \;=\; \gamma(0) - \hat{\mathbf{a}}^\top \boldsymbol{\gamma}_n(h).$$

Both depend only on $\gamma$, which the ARMA fit from Step 2 supplies via $\hat{\boldsymbol{\beta}}_{\mathrm{MLE}}$.

## 3.2 Implementation: $h$-step recursion

Inverting $\Gamma_n$ directly is impractical for $n = 2100$. For ARMA, the same innovations algorithm extends naturally to multi-step prediction. For $n > \max(p, q)$ and $h \geq 1$:

$$P_n Y_{n+h} \;=\; \sum_{i=1}^{p} \phi_i \, P_n Y_{n+h-i} \;+\; \sum_{j=h}^{q} \theta_{n+h-1,\,j} \, \big(Y_{n+h-j} - \hat{Y}_{n+h-j}\big),$$

with the convention $P_n Y_{n+h-i} = Y_{n+h-i}$ when $h - i \leq 0$. The MA-side sum vanishes once $h > q$, after which the forecast follows the AR recursion alone:

$$P_n Y_{n+h} \;=\; \sum_{i=1}^{p} \phi_i \, P_n Y_{n+h-i}, \qquad h > q.$$

For a causal AR this decays geometrically toward 0 (the process mean) as $h$ grows. The MSE $v_n(h)$ comes out of the recursion as a byproduct and satisfies

$$v_n(h) \;\xrightarrow{h \to \infty}\; \gamma(0) = \mathrm{Var}(Y_t),$$

reflecting that the long-horizon forecast carries no information beyond the unconditional mean.

### 3.3 Prediction intervals

Under Gaussian innovations, $Y_{n+h} - P_n Y_{n+h}$ is approximately $\mathcal{N}(0, v_n(h))$. An approximate 95% prediction interval on the residual scale is

$$\boxed{P_n Y_{n+h} \;\pm\; 1.96\,\sqrt{v_n(h)}.}$$

### 3.4 Inverse transforms back to MW

For $h = 1, \ldots, H$ extrapolate the deterministic cleaning components to time $n + h$.

Trend (with the chosen degree, say quadratic):

$$\hat{m}_{n+h} \;=\; \hat{\beta}_0 + \hat{\beta}_1(n+h) + \hat{\beta}_2(n+h)^2.$$

Yearly seasonal:

$$\hat{s}^{(\text{year})}_{n+h} \;=\; \sum_{j=1}^{k_y} \Big[\, \hat{a}^{(y)}_j \cos\big(\lambda^{(y)}_j (n+h)\big) + \hat{b}^{(y)}_j \sin\big(\lambda^{(y)}_j (n+h)\big) \,\Big].$$

Weekly seasonal:

$$\hat{s}^{(\text{week})}_{n+h} \;=\; \sum_{j=1}^{k_w} \Big[\, \hat{a}^{(w)}_j \cos\big(\lambda^{(w)}_j (n+h)\big) + \hat{b}^{(w)}_j \sin\big(\lambda^{(w)}_j (n+h)\big) \,\Big].$$

Combined additive future component:

$$F_{n+h} \;:=\; \hat{m}_{n+h} \;+\; \hat{s}^{(\text{year})}_{n+h} \;+\; \hat{s}^{(\text{week})}_{n+h}.$$

Point forecast and bounds in MW:

$$\begin{aligned}
\hat{X}_{n+h} \;&=\; \exp(\, P_n Y_{n+h} \;+\; F_{n+h}\,),\\
L^{(95)}_{n+h} \;&=\; \exp\Big(\, P_n Y_{n+h} - 1.96\sqrt{v_n(h)} \;+\; F_{n+h}\,\Big),\\
U^{(95)}_{n+h} \;&=\; \exp\Big(\, P_n Y_{n+h} + 1.96\sqrt{v_n(h)} \;+\; F_{n+h}\,\Big).
\end{aligned}$$

Because exp is nonlinear, the MW interval is asymmetric (log-normal).

### 3.5 Evaluation on a held-out test set

Hold out the last $H = 30$ observations as test data $\{\hat{x}_{n-H+j}\}_{j=1}^{H}$. Refit the chosen ARMA on the training portion only (residuals indexed $1, \ldots, n-H$) to obtain $\hat{\boldsymbol{\beta}}^{(\text{train})}_{\text{MLE}}$. Compute the forecasts $\hat{X}_{(n-H)+h}$ and intervals as above for $h = 1, \ldots, H$.

Metrics:

$$\text{RMSE} \;=\; \sqrt{\frac{1}{H}\sum_{h=1}^{H} \big(\hat{X}_{(n-H)+h} - \hat{x}_{(n-H)+h}\big)^2},$$

$$\text{MAE} \;=\; \frac{1}{H}\sum_{h=1}^{H} \big|\hat{X}_{(n-H)+h} - \hat{x}_{(n-H)+h}\big|,$$

$$\text{Coverage} \;=\; \frac{1}{H}\sum_{h=1}^{H} \mathbb{1}\Big[\, \hat{x}_{(n-H)+h} \in \big[L^{(95)}_{(n-H)+h},\, U^{(95)}_{(n-H)+h}\big] \,\Big].$$

Coverage should be close to 0.95.

## Summary of dependencies

$$\underbrace{X_t}_{\text{raw}} \xrightarrow{\log} \log X_t \xrightarrow{-\hat{s}_t^{(y)},\, -\hat{s}_t^{(w)},\, -\hat{m}_t} \underbrace{\hat{Y}_t}_{\text{Step 1 output}}$$

$$\underbrace{\hat{Y}_t}_{\text{Step 2 input}} \xrightarrow{\text{ID, MLE, AICC, diagnostics}} \underbrace{\hat{\boldsymbol{\beta}}_{\text{MLE}}}_{\text{Step 2 output}} \xrightarrow{\text{ARMA closed form}} \hat{\gamma}(\cdot)$$

$$\underbrace{\hat{\gamma}(\cdot)}_{\text{Step 3 input}} \xrightarrow{\text{innovations algo / } h\text{-step recursion}} \left\{P_n Y_{n+h},\, v_n(h)\right\}_{h=1}^{H} \xrightarrow{+F_{n+h},\, \exp} \underbrace{\left\{\hat{X}_{n+h},\, L_{n+h}^{(95)},\, U_{n+h}^{(95)}\right\}_{h=1}^{H}}_{\text{Step 3 output}}.$$